

## Assignment 2

### Date: 28th Oct, 2025

**Instructions:** Perform a complete dry run of the program on paper, take clear pictures of the working steps, and compile them into a PDF file for assignment submission.

#### Q#01

A retail company stores the daily sales amounts of each branch.

To find the **average daily sales**, the company's report generator recursively processes each day's sale value and computes the mean.

This helps in understanding performance trends without using loops. Write a recursive formulation, c++ code and recurrence relation and calculate the average daily sales.

#### Q#02

Solve the recurrence relation from the above given problem to find the time complexity of the code.

#### Q#03

An online shopping platform stores its product IDs in a sorted array for efficient searching.

When a customer searches for a specific product, the system must quickly locate it using recursive binary search.

You are required to write a recursive function in C++ that searches for a given product ID in the sorted list of product IDs and returns the index (position) if found, or **-1** if the product does not exist in the inventory.

#### Q#04

Solve the recurrence relation from the above given problem to find the time complexity of the code of Binary Search